

Frank — AMC Drill M8 · logic / invariant

25 Jun 2026 | ANSWER KEY & method — for the parent

1. Answer: **300** [M8 · twin of real 2018 Q24 · 5 marks]

Working: 4 odd corners → fence total $2 \times 40 + 2 \times 30 + 2 \times 50 = 240$; re-walk the two shortest sides = $2 \times 30 = 60$; route = $240 + 60 = 300$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

2. Answer: **192** [M8 · twin of real 2018 Q24 · 5 marks]

Working: 4 odd corners → fence total $2 \times 30 + 2 \times 16 + 2 \times 34 = 160$; re-walk the two shortest sides = $2 \times 16 = 32$; route = $160 + 32 = 192$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

3. Answer: **140** [M8 · twin of real 2018 Q24 · 5 marks]

Working: 4 odd corners → fence total $2 \times 24 + 2 \times 10 + 2 \times 26 = 120$; re-walk the two shortest sides = $2 \times 10 = 20$; route = $120 + 20 = 140$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

4. Answer: **433** [M8 · twin of real 2018 Q27 · 7 marks]

Working: the hidden (opposite) faces are $433 - 100 = 333$, $433 - 130 = 303$, $433 - 133 = 300$ — all use only the allowed digits and are different, so the largest total is **433**.

★ List the allowed numbers; for each candidate total T, check whether all three hidden faces (T minus each visible number) are themselves allowed and all different. Take the largest T that works.

Trap: assuming the three visible faces can be opposite each other — they meet at a corner, so they are adjacent; their opposites are the three hidden faces.

5. Answer: **535** [M8 · twin of real 2018 Q27 · 7 marks]

Working: the hidden (opposite) faces are $535 - 111 = 424$, $535 - 121 = 414$, $535 - 124 = 411$ — all use only the allowed digits and are different, so the largest total is **535**.

★ List the allowed numbers; for each candidate total T, check whether all three hidden faces (T minus each visible number) are themselves allowed and all different. Take the largest T that works.

Trap: assuming the three visible faces can be opposite each other — they meet at a corner, so they are adjacent; their opposites are the three hidden faces.

6. Answer: **563** [M8 · twin of real 2018 Q27 · 7 marks]

Working: the hidden (opposite) faces are $563 - 121 = 442$, $563 - 122 = 441$, $563 - 141 = 422$ — all use only the allowed digits and are different, so the largest total is **563**.

★ List the allowed numbers; for each candidate total T, check whether all three hidden faces (T minus each visible number) are themselves allowed and all different. Take the largest T that works.

Trap: assuming the three visible faces can be opposite each other — they meet at a corner, so they are adjacent; their opposites are the three hidden faces.